



**Kouhbanan organic biochar mine .Kerman.
IRAN.**

**بیوچار طبیعی معدنی کوهبنان کرمان .
ایران .**

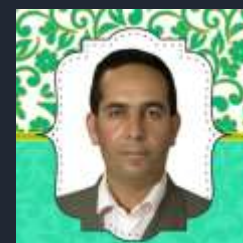
**Eng, Mr.Mohammad Majidi Kouhbanani
آقای مهندس محمد مجیدی کوه بنانی**

In the name of Allah

Delete Petroleum hydrocarbons and heavy metals from contaminated soils using Kerman natural mineral biochar

Soil pollution is one of the important environmental problems caused by the use of Kerman natural mineral biochar able is to

Kerman natural mineral biochar Using materials _
Biological and its unique structure can Improve soil
physico-chemistry and remove soil pollutants,
including petroleum hydrocarbons and heavy metals.



**By Mr.
Engineer Mohammad Majidi KouhBanani**

The importance of removing petroleum hydrocarbons and heavy metals from contaminated soils



environmental effects

Contaminated soils have serious environmental and health risks and this issue is eliminated Effectively, petroleum hydrocarbons and heavy metals are necessary

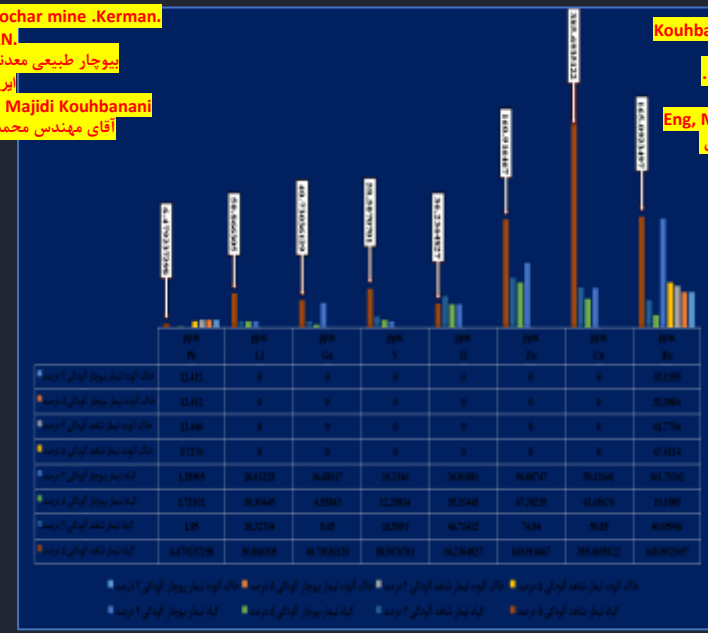
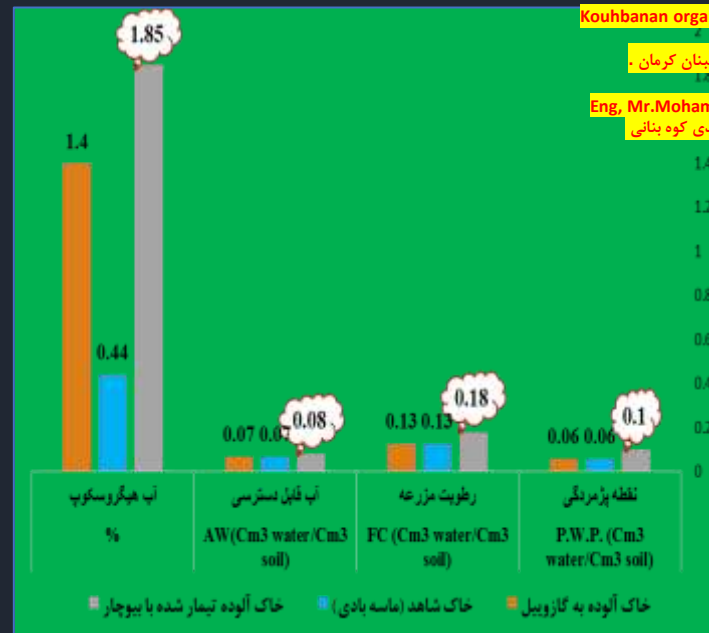
Potential solutions

The development of efficient methods to remove these pollutants is necessary to protect the ecosystem and human well-being

Long-term benefits

Remediation of contaminated soils leads to improvement Efficiency Agriculture and environment healthier _ For future generations to be

Removal of petroleum hydrocarbons and heavy metals using Kerman natural mineral biochar



Improve soil quality

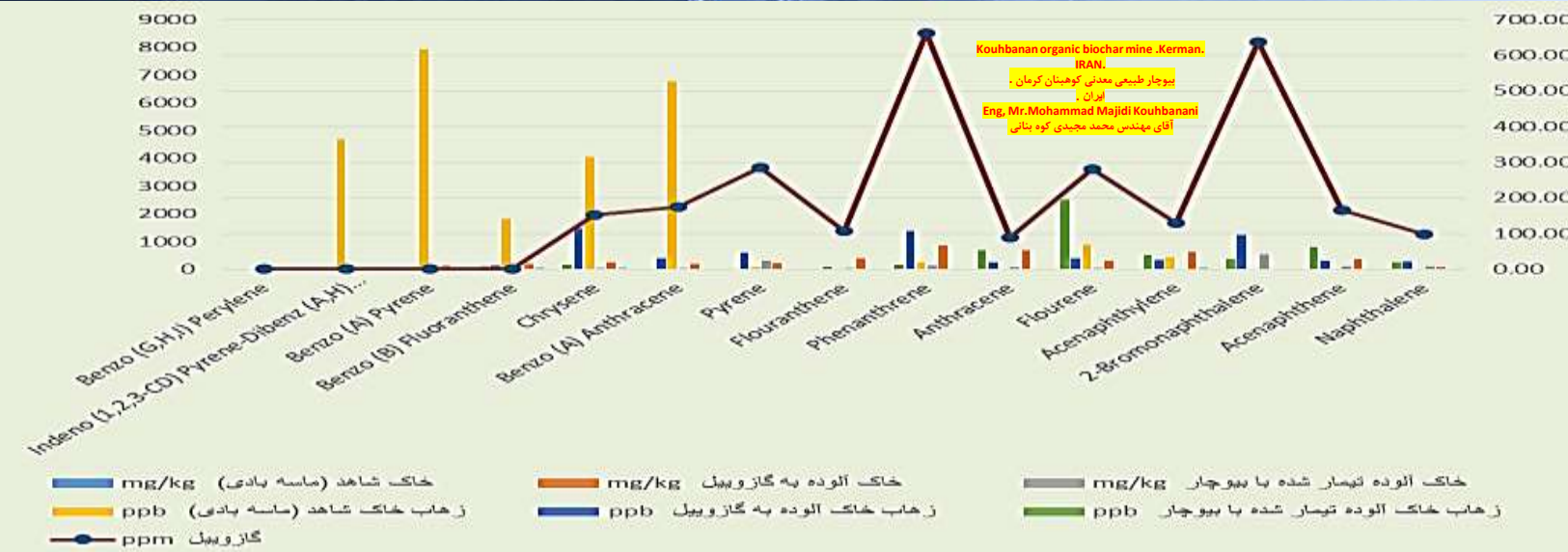
Kerman natural mineral biochar
It has the ability to improve the physical and chemical properties of contaminated soils and creates a more favorable environment for the growth of plants

Reduce toxicity

using _ Kerman natural mineral biochar
It helps in reducing the bioavailability and toxicity of heavy metals, reducing their adverse effects on plants and organisms

Less environmental impact

using _ Kerman natural mineral biochar
For remedial purposes, it prevents the leakage and penetration of petroleum hydrocarbons and heavy metals into underground and surface water.



The process of removing hydrocarbons and petroleum heavy metals using Kerman natural mineral biochar

1

Analysis of soil

Conducting a complete soil test to identify the types and levels of petroleum hydrocarbons and heavy metals in the contaminated area

2

Application Kerman natural mineral biochar

Appropriate amounts of Kerman natural mineral biochar Apply to contaminated soil to ensure proper mixing and distribution for optimal remediation effectiveness

3

Monitoring and Evaluation

To make the necessary adjustments and ensure successful results, regularly review the progress and effectiveness of the based modification process Kerman natural mineral biochar Check out

Improving the physicochemical properties of contaminated soils Using natural mineral biochar from the mountains of Kerman

1 Modification of soil structure

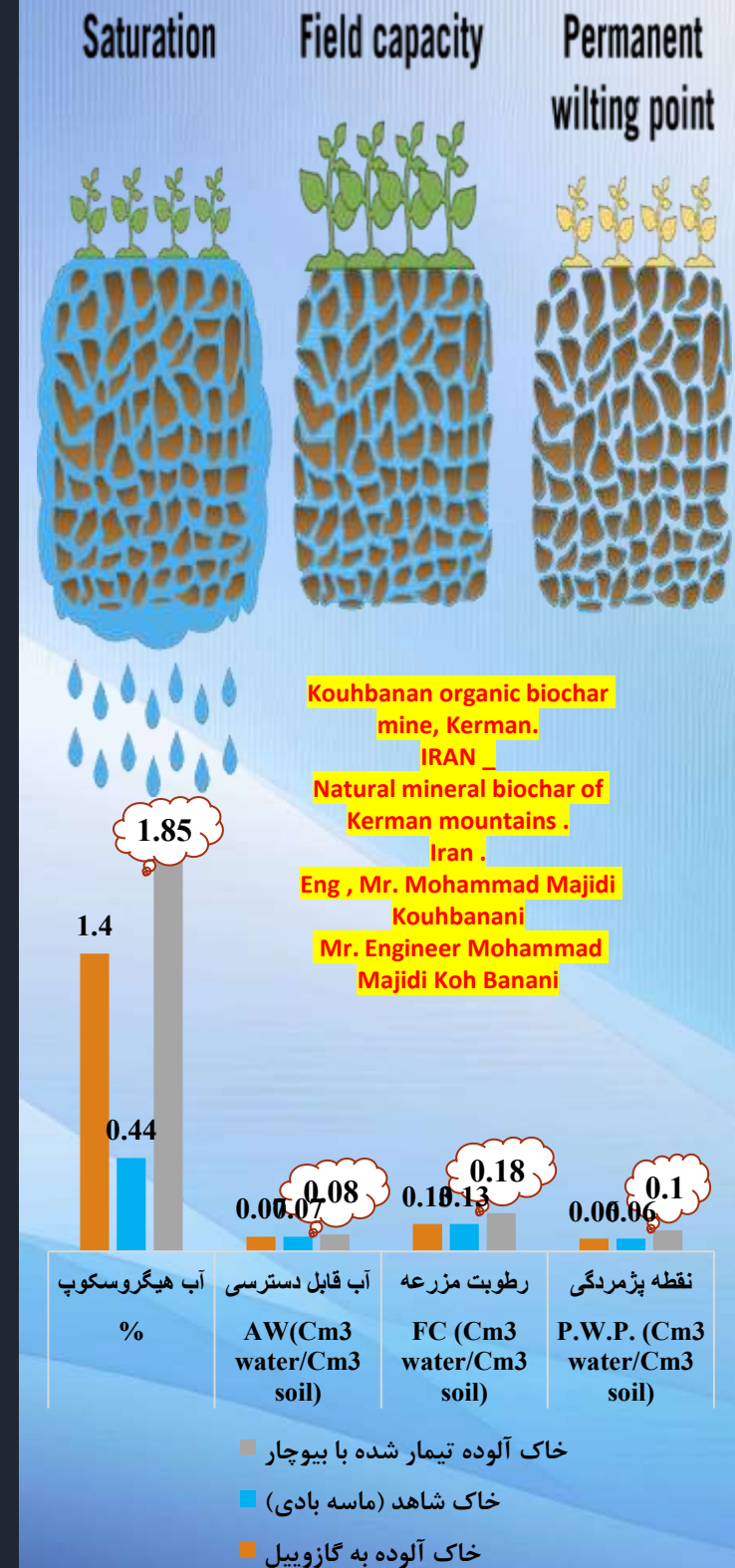
Kerman natural mineral biochar It improves the soil texture, porosity and water retention, which leads to the improvement of the overall soil structure and fertility

3 Optimal nutrient availability

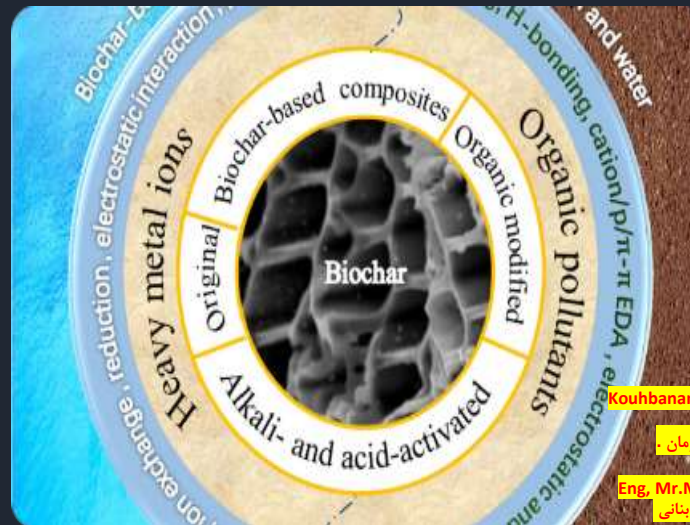
Kerman natural mineral biochar The ability to retain and release nutrients gradually increases the availability of essential elements for plant growth in contaminated soils

2 Reduce erosion

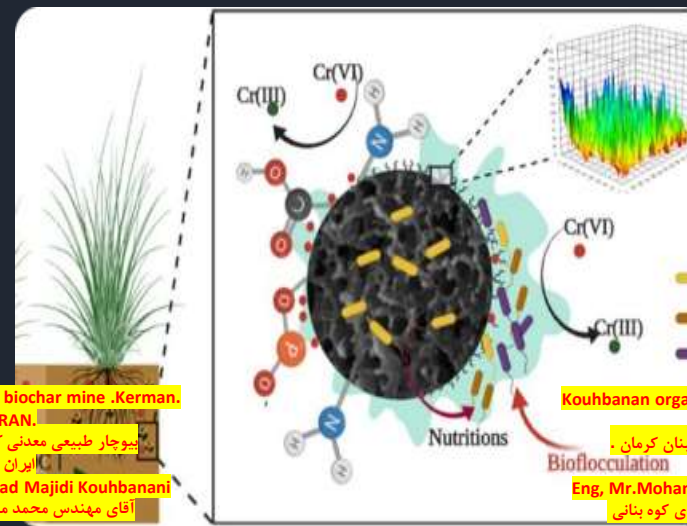
Composition Kerman natural mineral biochar It helps to prevent soil erosion and nutrient leaching and leads to sustainable soil use it becomes



Eliminate and reduce provision biological Metals Heavy At Soil Hi infected With Application Kerman natural mineral biochar



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Complex mechanism

Kerman natural mineral biochar It can immobilize heavy metals through complexation, reducing their availability for plant uptake and minimizing potential toxicity.

Stabilization effect

Add Kerman natural mineral biochar It stabilizes heavy metals and prevents them from leaching into water sources, reducing environmental hazards

Long term correction

Kerman natural mineral biochar Facilitates long-term remediation of soils contaminated with heavy metals and provides sustainable and environmentally friendly solutions

Removal of aromatic hydrocarbons from contaminated soils using Kerman natural mineral biochar

Absorption capacity

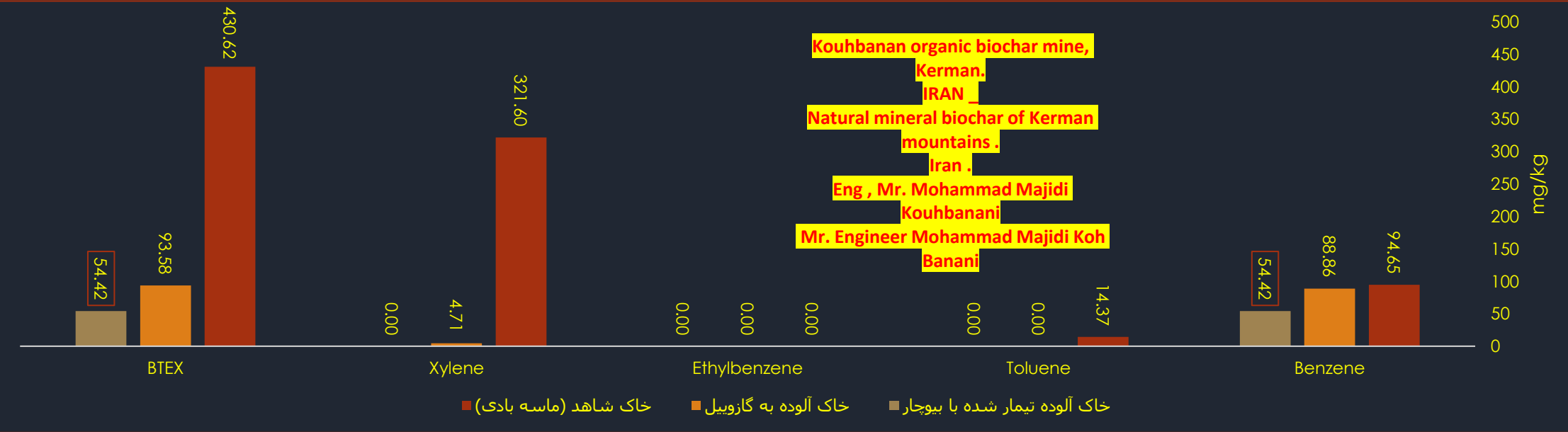
high level Kerman natural mineral biochar And the porosity enables the effective absorption of aromatic hydrocarbons and reduces their concentration in the soil.

Parsing and converting

Biochar develops the breakdown and conversion of aromatic hydrocarbons and enhances natural remediation processes in contaminated soils.

soil regeneration

using _ Kerman natural mineral biochar By increasing microbial activity, it regenerates contaminated soils and helps to break down aromatic hydrocarbons.



Removal of persistent organic compounds from contaminated soils using Kerman natural mineral biochar

1

soil amendment with Kerman natural mineral biochar

Application Kerman natural mineral biochar
As a soil conditioner, it causes physical and chemical interactions that lead to the destruction, stabilization and immobilization of stable organic compounds.

2

Soil integration

Biochar integrated with contaminated soils creates favorable environmental conditions by increasing the contact surface, which leads to the decomposition of stable organic compounds.

3

soil regeneration

Enhanced microbial activity with The natural mineral biochar of Kerman mountain accelerates the degradation of stable organic compounds and helps to improve the soil effectively.

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Removal of aromatic hydrocarbons and petroleum hydrocarbons from contaminated soils using Kerman natural mineral biochar

Interaction of compounds

Kerman natural mineral biochar In interaction with aromatic and petroleum hydrocarbons, reduce their mobility and increase their decomposition in polluted soils.

Environmental Protection

using _ Kerman natural mineral biochar It helps protect the environment by preventing the washing and spreading of aromatic and petroleum hydrocarbons

Stable modification

based modification Kerman natural mineral biochar It provides a sustainable and environmentally friendly approach to the effective removal of hydrocarbon pollutants

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